

Software Development Life Cycle (SDLC) Cheat Sheet

1. SDLC Overview:

SDLC is the structured process of developing software from idea to deployment and maintenance. It ensures quality, efficiency, and predictability.

Phases of SDLC:

- ➔ Requirement Gathering – Understanding what needs to be built.
- ➔ Planning – Estimating time, resources, technologies.
- ➔ Design – System architecture, data flow, UI/UX planning.
- ➔ Development – Actual coding of the system.
- ➔ Testing – Verifying functionality and fixing bugs.
- ➔ Deployment – Pushing the application to production.
- ➔ Maintenance – Updates, patches, improvements.

SDLC Flow (Text Diagram):

Requirement → Planning → Design → Development → Testing → Deployment → Maintenance

(Loop repeats for next features in Agile development.)

Waterfall vs Agile SDLC:

Waterfall:

- Linear, step-by-step.
- Requirements fixed before development.
- Changes are difficult once development starts.

Agile:

- Iterative and incremental.
- Uses Scrum boards, sprints, Jira.
- Requirements evolve with feedback.

SDLC Example for Your RAG Project:

Requirement: Build an AI system that retrieves domain knowledge using LLM + embeddings.

Planning: Choose Python, Flask, FAISS/Chroma DB, OpenAI/Llama models.

Design: Retriever → LLM → Post-processor architecture.

Development: Write Flask APIs, embedding pipeline, vector DB integration.

Testing: Check retrieval accuracy, API responses, edge cases.

Deployment: Host API on local server or cloud.

Maintenance: Improve retrieval, add new documents, optimize latency.